UNIVERSITY OF THE PEOPLE

MATH 1280-01 - AY2026-T4

Assignment Activity Unit 6

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Geometric and Hypergeometric Probability Distributions

Probability distributions are important tools in statistics because they help organizations analyze uncertainty and make better decisions. In a hospital emergency room, probability models can help improve patient flow, reduce waiting times, and optimize the use of medical resources. This assignment applies the geometric and hypergeometric distributions to analyze hospital operations.

Geometric Distribution

The geometric distribution is used when we want to determine the probability that the first success occurs on a specific trial. In this scenario, the hospital is testing the effectiveness of its triage process. The probability of successfully triaging a patient on the first attempt is:

$p=0.7p = 0.7$p=0.7

The probability of failure is:

$q=1-pq = 1 - p$ q=1−p $q=1-0.7=0.3q = 1 - 0.7 = 0.3$q=1−0.7=0.3

We are asked to find the probability that the third triage attempt will be the first successful one.

The geometric probability formula is:

$$P(X = x) = (1 - p)^{x-1}p$$

Where:

- $x = 3$
- $p = 0.7$
- $1 - p = 0.3$

Substituting the values into the formula:

$$P(X = 3) = (0.3)^{3-1}(0.7)$$

$$P(X = 3) = (0.3)^{2}(0.7)$$

$$P(X = 3) = (0.09)(0.7)$$

$$P(X = 3) = 0.063$$

Therefore, the probability that the third triage attempt will be the first successful one is:

0.063

or 6.3%.

This means that there is a relatively small probability that two unsuccessful triage attempts will occur before the first successful triage.

Hypergeometric Distribution

The hypergeometric distribution is used when sampling is done without replacement from two groups. In this case, the hospital has 20 recently arrived patients, and 5 of them have flu. A sample of 4 patients is selected randomly for testing. We want to find the probability that exactly 2 of the selected patients have flu.

Let:

- Total population: $N=20$
- Number with flu: $K=5$
- Sample size: $n=4$
- Desired successes: $x=2$

The hypergeometric probability formula is:

$$P(X = x) = \frac{\binom{K}{x}\binom{N-K}{n-x}}{\binom{N}{n}}$$

Substituting the values:

$$P(X = 2) = \frac{\binom{5}{2}\binom{15}{2}}{\binom{20}{4}}$$

Now calculate each combination:

$$\binom{5}{2} = 10$$

$$\binom{15}{2} = 105$$

$$\binom{20}{4} = 4845$$

Substitute these values:

$$P(X = 2) = \frac{(10)(105)}{4845}$$

$$P(X = 2) = \frac{1050}{4845}$$

$$P(X = 2) \approx 0.2167$$

Therefore, the probability that exactly 2 of the 4 selected patients have flu is:

0.2167

or approximately 21.67%.

This result shows that there is a moderate probability of selecting exactly two flu patients from the sample.

Conclusion

Both geometric and hypergeometric distributions are useful statistical tools in healthcare operations. The geometric distribution helps hospitals analyze repeated attempts until the first successful outcome occurs, such as successful patient triage. The hypergeometric distribution helps evaluate probabilities when selecting samples without replacement, such as identifying flu cases among patients. These probability models support better planning, decision-making, and resource management in hospital environments.

References

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